

Summary: A Neural Algorithm of Artistic Style

Gatys, Ecker & Bethge (2015)

1 Overview

Gatys, Ecker, and Bethge show that a Convolutional Neural Network (CNN) trained for object recognition (VGG-19) implicitly separates an image’s *content* (the objects and their spatial arrangement) from its *style* (texture, color, and local structure). By defining two independent loss functions over the network’s feature activations, one for content and one for style, they synthesize new images that combine the content of one photograph with the style of an artwork. This report summarizes the mathematical formulation underlying the method.

2 Feature Representations

An input image $\vec{x}$ is passed through the VGG-19 network (only the 16 convolutional and 5 pooling layers are used; fully connected layers are discarded, and max-pooling is replaced by average-pooling for smoother gradients). At layer l , which contains N_l filters, each producing a feature map of size M_l (height $\times$ width), the activations are stored as a matrix

$$F^l \in \mathbb{R}^{N_l \times M_l}, \quad F_{ij}^l = \text{activation of filter } i \text{ at spatial position } j. \quad (1)$$

2.1 Content Representation

The raw filter activations F^l at a chosen layer l are used directly as the *content representation*: higher layers preserve high-level object arrangement but lose exact pixel detail, while lower layers reconstruct the image almost perfectly.

2.2 Style Representation

Style is captured by feature *correlations* across the spatial extent of each layer, computed as a Gram matrix

$$G^l \in \mathbb{R}^{N_l \times N_l}, \quad G_{ij}^l = \sum_k F_{ik}^l F_{jk}^l. \quad (2)$$

Because G^l discards spatial position and only encodes co-occurrence statistics between filters, it yields a stationary, multi-scale texture descriptor. Combining G^l across several layers gives the full style representation.

2.3 Why Style is Captured by the Gram Matrix

Prior to this novel idea by Gatys, in the year 2000, texture synthesis was usually done by passing a bunch of linear filters over an image and then using some linear combination of these filters to generate an image of the same texture. Here, we see that style is done by using gram matrix.

$$G_{ij} = \sum_{k=1}^M F_{ik} F_{jk} \quad (3)$$

the reason is two core behaviors- **spatial averaging** and **feature co-occurrence**. Spatial averaging means that the gram matrix's values don't care where in space two filters correspond to each other. The more the value of the correlation between activations, the more the value in the gram matrix irrespective of the location. Feature co-occurrence refers to when two features such as -cobalt blue and diagonal edges- correspond to each other, then this specific texture is prevalent thus part of the style of the image.

3 Loss Functions

3.1 Content Loss

Let $\vec{p}$ be the original (content) image and $\vec{x}$ the image being synthesized, with feature representations P^l and F^l at layer l . The content loss is the squared error between activations:

$$\mathcal{L}_{\text{content}}(\vec{p}, \vec{x}, l) = \frac{1}{2} \sum_{i,j} \left(F_{ij}^l - P_{ij}^l \right)^2. \quad (4)$$

Its gradient with respect to the layer activations (passed back through ReLU) is

$$\frac{\partial \mathcal{L}_{\text{content}}}{\partial F_{ij}^l} = \begin{cases} (F^l - P^l)_{ij} & \text{if } F_{ij}^l > 0 \\ 0 & \text{if } F_{ij}^l < 0. \end{cases} \quad (5)$$

3.2 Style Loss

Let $\vec{a}$ be the artwork (style) image, with Gram-matrix representations A^l and G^l for the artwork and the generated image, respectively. Each layer's contribution to the style loss is a normalized squared error between Gram matrices:

$$E_l = \frac{1}{4N_l^2 M_l^2} \sum_{i,j} \left(G_{ij}^l - A_{ij}^l \right)^2. \quad (6)$$

The total style loss is a weighted sum over a chosen set of layers:

$$\mathcal{L}_{\text{style}}(\vec{a}, \vec{x}) = \sum_{l=0}^L w_l E_l, \quad (7)$$

where w_l are per-layer weights (typically $1/(\text{number of active layers})$, and 0 elsewhere).

Each entry of the Gram matrix is a sum over the M_l spatial positions of a layer:

$$G_{ij}^l = \sum_{k=1}^{M_l} F_{ik}^l F_{jk}^l. \quad (8)$$

Since this is a sum of M_l terms, the typical magnitude of G_{ij}^l scales roughly linearly with M_l : a layer with a larger feature map (more spatial positions) produces larger raw Gram matrix values, purely as a consequence of summing over more terms, independent of how strong the underlying style signal actually is. The same is true of A_{ij}^l , the Gram matrix of the target style image. Consequently, the difference between corresponding entries, $G_{ij}^l - A_{ij}^l$, also scales roughly linearly with M_l , since both terms individually scale with M_l . Therefore, the style loss must scale with M_l^2 . Dividing by M_l^2 cancels this inflation, so that layers with small feature maps (deep layers) and layers with large feature maps (shallow layers) contribute to the loss on a comparable scale. The remaining factor of $\frac{1}{N_l^2}$ separately averages over the $N_l \times N_l$ entries of the Gram matrix itself, and the factor of $\frac{1}{4}$ is a convenient constant that simplifies the derivative in Equation (6).

The corresponding gradient is

$$\frac{\partial E_l}{\partial F_{ij}^l} = \begin{cases} \frac{1}{N_l^2 M_l^2} ((F^l)^T (G^l - A^l))_{ji} & \text{if } F_{ij}^l > 0 \\ 0 & \text{if } F_{ij}^l < 0. \end{cases} \quad (9)$$

3.3 Total Loss

The synthesized image is obtained by jointly minimizing a weighted combination of the two losses:

$$\mathcal{L}_{\text{total}}(\vec{p}, \vec{a}, \vec{x}) = \alpha \mathcal{L}_{\text{content}}(\vec{p}, \vec{x}) + \beta \mathcal{L}_{\text{style}}(\vec{a}, \vec{x}), \quad (10)$$

where α and β control the relative emphasis on content versus style. In the paper’s experiments, content is matched at layer `conv4_2`, style is matched across `conv1_1` through `conv5_1` (equal weights), and the ratio α/β is set to 10^{-3} or 10^{-4} depending on the desired stylization strength.

4 Optimization Procedure

Image synthesis begins from a white-noise image $\vec{x}$. Gradients of $\mathcal{L}_{\text{total}}$ with respect to the pixels of $\vec{x}$ are computed via standard backpropagation through the fixed, pre-trained VGG-19 network. Only the image is updated; the network weights are frozen. Gradient descent is then performed on $\vec{x}$ until it simultaneously satisfies the content constraint (matching activations at the chosen layer) and the style constraint (matching Gram matrices across the chosen layers).

5 Key Insight

Because content and style are represented by different mathematical objects, raw activations F^l versus feature correlations G^l , they can be optimized largely independently. Increasing α/β favors content fidelity; decreasing it favors style fidelity. Including style terms from progressively higher layers captures texture structure at increasingly large spatial scales, since receptive field size and feature complexity grow with depth.